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4. The table shows various stages in the development of the Earth's atmosphere since its formation 4 500 million years ago.

Stage	Major events	Gases present in the atmosphere
1	volcanic eruptions	carbon dioxide, water vapour, methane, ammonia
2	oceans form	carbon dioxide, methane, ammonia
3	green plants evolve	carbon dioxide, nitrogen, oxygen
4	most carbon dioxide becomes locked in rock and fossil fuels	nitrogen, oxygen, water vapour

(a) Which **one** of these statements best describes how the oceans were formed?

Put a tick (✓) in the box next to the correct answer. [1]

Water vapour evaporated to form clouds ☐

The Earth cooled so water vapour condensed ☐

Bacteria and algae turned the water vapour into liquid water ☐

There were no more volcanoes to produce water vapour ☐

(b) Explain why the appearance of green plants was an important stage in the development of the atmosphere. [2]

....

....

....

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(c) The atmosphere today contains nitrogen, oxygen and water vapour. [1]

Use the following information to identify another gas present.

- a Group 0 gas
- the third most abundant in the atmosphere
- used in light bulbs and as an inert atmosphere for welding

....

(d) Nitrogen can be obtained by heating sodium azide, NaN_3. Sodium is also produced in the reaction.

Complete the balancing of the equation for this reaction. [1]

$$2NaN_3 \longrightarrow 2Na + \boxed{}\,N_2$$

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